A. **Multiplicative Functions.** We will show $\tau$ and $\sigma$ both have a special property.

Let $f$ be an arithmetic function. Then $f$ is **multiplicative** if and only if both

**Theorem.** $\tau$ and $\sigma$ are multiplicative.

**Proof.** First it is clear that:

Now suppose that $m$ and $n$ are relatively prime.

**Lemma.** If $m \perp n$, the divisors $d$ of $mn$ may be uniquely decomposed as

$d =$

**Proof.** If $m$ or $n$ are $1$, this is obvious. So assume $m$ and $n$ are greater than $1$.

Take their prime decompositions

$m =$

$n =$

where, because $m \perp n$, none of the:

Then, the prime decomposition of $mn$ is

$mn =$

and thus each divisor of $mn$ is uniquely of the form:

$d =$

which uniquely specifies $d_1$ and $d_2$.

□

By the lemma, since $m \perp n$, the number of divisors of $mn$ is:

Hence: $\tau(mn) =$

Again by the lemma, since $m \perp n$, the sum of divisors of $mn$ is

We established this property of double sums last class.

Hence: $\sigma(mn) =$

□

B. **Powers of Primes.** Multiplicative functions are determined at powers of primes.

**Theorem.** Let $f$ be a multiplicative function.

Then $f$ is uniquely determined by its values at positive powers of primes.

Specifically, if $n$ is an integer greater than $1$ with prime decomposition

$n =$

Then, since:

$f(n) =$

Using this principle, we may readily provide formulas for the $\tau$ and $\sigma$ functions.

**Corollary.** Let $n$ be a positive integer with prime decomposition

$n = p_1^{k_1} \cdots p_r^{k_r}$

Then, since

$\tau(p^k) =$

$\sigma(p^k) =$

we have:

$\tau(n) =$

$\sigma(n) =$

For example we use $360 =$

$\tau(360) =$

$\sigma(360) =$

**Example 1.** Prove that the following functions are multiplicative.

- The constant $1$ function defined by $\mathbb{1}(n) = 1$ for all $n$.
- The **identity function** $\mathrm{id}$ defined by $\mathrm{id}(n) = n$ for all $n$.

**Proof.** In fact they are **completely multiplicative**, which for an arithmetic $f$ means

as we can see by letting positive integers $m$ and $n$ be given and computing

□

**Example 2.** Let $a$ be a nonzero integer. Prove that the function $\gcd_a$ defined by

$\gcd_a(n) =$

is multiplicative.

Hence, if $m \perp n$, then $\gcd(a, mn) =$

**Proof.** First note that

$\gcd_a(1) =$

Next let positive integers $m$ and $n$ be given such that:

Because $m \perp n$, the divisors of $mn$ are exactly of the form

By "exactly" of this form, we mean that each divisor of $mn$ has this form, and any integer has this form is a divisor of $mn$.

$d =$

where since $m \perp n$ it follows that:

Among these divisors of $mn$, those that also divide $a$ are exactly of the form

By "exactly" of this form, we mean that each common divisor of $mn$ and $a$ has this form, and any integer that has this form is a common divisor of $a$ and $mn$.

$d =$

because $d_1 \perp d_2$ and $d_1 \mid a$ and $d_2 \mid a$ implies:

Therefore, the greatest common divisor of $mn$ and $a$ is

In other words

$\gcd_a(mn) =$

□